

3.12 Technical Tools and Software

3.12.1 Data Processing Tools

The researcher used Pandas and NumPy for data manipulation and modeling, respectively.

Pandas was used to conduct the cleaning and manipulation activity through its DataFrames.

NumPy managed numerical computation and multi-dimensional arrays management. Scikit-learn implemented the development and training of machine learning models by providing a wide range of algorithms, along with associated tools to conduct evaluation and selection. It also combined these libraries in the pipeline for smooth processing and model development.

Visualization of data in readable and informative plots and charts was done using Matplotlib and Seaborn. Visualization with Matplotlib included all static, animations, and interactive visualizations, while complex visualizations with Seaborn were built on top of Matplotlib in order to portray the trending aspect in the data, model performance, and other metrics. The line graphs and bar charts were developed using Matplotlib; Seaborn will build up the heatmaps and scatter plots, depicting correlations of data and its pattern. This was then integrated onto the UI for concise insight into milk quality trends and model performance.

3.12.2 Machine Learning Frameworks

Scikit-learn was used to implement the main classification models considered in the study, including Random Forest, Support Vector Machine, and XGBoost. The library provided efficient tools for preprocessing, model training, hyperparameter tuning, and performance evaluation.

PyTorch supported the comparative Artificial Neural Network implementation, while joblib and related Python utilities were used for saving and loading trained models during prototype integration. Together, these frameworks enabled both experimental comparison and practical embedding of the selected classifier in the desktop interface.